AQ700 method

EPA-165-E Rev. A

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# Sulfate in Drinking, Saline and Surface Waters, and Domestic and Industrial Wastes

## 1 SCOPE

This method covers the determination of sulfate in drinking, surface, and saline waters, and domestic and industrial wastes.

This method is approved for the Clean Water Act for use in wastewater compliance monitoring, under National Pollutant Discharge Elimination System (NPDES) and may be recommended for use in NSDWR (National Secondary Drinking Water Regulations) compliance monitoring. The laboratory should consult the most recent Code of Federal Regulations for applicability and approval by the Clean Water and Safe Drinking Water Acts (40 CFR 136 and 141).

## 2 RANGE OF APPLICATION

| | |
|---|---|
| Range | 2.0 – 40 mg $SO_4$/L |

## 3 METHOD DETECTION LIMIT

| | |
|---|---|
| By EPA Procedure | MDL = 0.1 mg $SO_4$/L |

## 4 METHOD PRINCIPLE

Sulfate ion is converted to a barium suspension under controlled conditions. The resulting turbidity is determined using a filter photometer at 405 nm.

## 5 REFERENCES

ASTM D516-16, *Standard Test Method for Sulfate Ion in Water,* ASTM International, West Conshohocken, PA, 2016, www.astm.org

ISO/DIS 15923-1, *Water Quality - Determination of selected parameters by a discrete analysis system - Part 1: Ammonium, nitrate, nitrite, chloride, orthophosphate, sulfate and silicate with photometric detection.*

## 6 INTERFERENCES

As much as 50 mg $Fe^{3+}$/L, 10 mg $Cu^{2+}$/L, and 10 mg $SiO_2$/L can be tolerated. High silica concentrations cause positive interference. High iron concentrations can cause precipitation of, and subsequent loss of, phosphorus.

Insoluble suspended matter in the sample must be removed. Dark colors that cannot be compensated for in the procedure interfere with the measurement of suspended barium sulfate ($BaSO_4$).

Polyphosphates as low as 1.0 mg/L will inhibit barium sulfate precipitation causing a negative interference. Phosphonates present in low concentrations, depending on the type of phosphonate, will also cause a negative interference.

Silica in excess of 500 mg/L may precipitate along with a barium sulfate causing a positive interference.

Chloride in excess of 5000 mg/L will cause a negative interference. Aluminum, polymers, and large quantities of organic material present in the test sample may cause the barium sulfate to precipitate nonuniformly.

In the presence of organic matter certain bacteria may reduce sulfate to sulfide. To minimize the action of sulfate reducing bacteria, samples should be refrigerated at 4$^{o}$C when the presence of such bacteria is suspected.

Bias from sample turbidity or color is correctable, using sample blanking feature of AQ software.

## 7 REAGENTS: CHEMICALS REQUIRED AND SAFETY INFORMATION

Consult the safety data sheets (SDS) for details on safe handling of chemicals.

| Chemical | Formula | C.A.S. No. | Safety Classification |
|---|---|---|---|
| Barium chloride, dihydrate | $BaCl_2 \cdot 2H_2O$ | 10326-27-9 | GHS05 |
| Gelatin | $C_6H_{12}O_6$ | 9000-70-08 | -- |
| Hydrochloric acid, concentrated | HCl | 7647-01-0 | GHS05, GHS07 |
| Sodium chloride | NaCl | 7647-14-5 | -- |
| Sodium sulfate | $Na_2SO_4$ | 7757-82-6 | -- |

| | | |
|---|---|---|
| GHS01: Danger – Explosive | GHS02: Danger – Flammable | GHS03: Danger – Oxidizing |
| GHS04: Warning – Compressed Gas | GHS05: Warning/Danger – Corrosive | GHS06: Danger – Toxic |
| GHS07: Warning – Irritant | GHS08: Danger – Health Hazard | GHS09: Warning/Danger – Aquatic Toxicity |

**Note:** Persons using this method should be familiar with normal laboratory practice. This method does not purport to address all of the safety problems, if any, associated with its use. It is the responsibility of the user to establish appropriate safety and health practices and to ensure compliance with local regulatory conditions.

## 8 REAGENT SOLUTIONS

Use reagent-grade chemicals, certified for analytical or general laboratory use. SEAL recommends using ACS Reagent Grade whenever possible. Reagents of technical or commodity grade must be validated by the user. Use high-purity reagent water, distilled or deionized, and free from organic contamination, therefore Grade 1 (ISO Standard 3696) or better than Type II (ASTM Standard D1193) waters are suitable.

### 8.1 HYDROCHLORIC ACID, 0.1 N

| | |
|---|---|
| Hydrochloric acid, concentrated | 2.1 mL |
| DI Water | dilute to 250 mL |

To approximately 200 mL of DI water, add 2.1 mL of hydrochloric acid, concentrated. Dilute to the mark with deionized water.

Store at room temperature. Stable for 6 months.

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### 8.2 TURBIDIMETRIC REAGENT

| | |
|---|---|
| Gelatin | 0.25 g |
| Barium chloride | 5.0 g |
| Sodium chloride | 10.0 g |
| Hydrochloric acid, 0.1 N (see above) | 17.52 mL |
| DI Water | dilute to 500 mL |

SEAL Analytical recommends purchasing Sigma-Aldrich catalog number G9391 gelatin from bovine skin powder (alternatives may be purchased with these specifications: gel strength ~225 bloom, Type B, or similar). In a beaker, combine 5.0 g of Barium Chloride (BaCl2), 10.0 g of Sodium Chloride (NaCl) and 0.25 g gelatin in 300 mL of deionized water. Carefully add 17.52 mL of 0.1N Hydrochloric acid and stir while gently heating for about 1 hour to dissolve. Dilute to 500 mL with deionized water after it is completely cool. Shake well before each use and best made the day before use.

Stable for 1 month.

## 9 STANDARD SOLUTIONS

### 9.1 SULFATE STOCK STANDARD SOLUTION (1000 mg $SO_4$/L)

| | |
|---|---|
| Sodium sulfate | 0.739 g |
| DI water | dilute to 500 mL |

To a 500 mL volumetric flask, dissolve 0.739 g sodium sulfate (dried at 105$^o$C) in a deionized water and dilute to the mark.

Store at 4$^o$C in a plastic bottle. Stable for 1 month.

### 9.2 AUTO CALIBRATION STANDARD – SULFATE (40 mg $SO_4$/L)

| | |
|---|---|
| Sulfate stock standard (1000 mg/L, above) | 8 mL |
| DI Water | dilute to 200 mL |

To a 200 mL volumetric flask, pipet 8 mL of sulfate stock standard (1000 mg $SO_4$/L) and dilute to the mark with deionized water. Invert to mix.

Store at 4$^o$C. Prepare fresh weekly.

### 9.3 MANUAL CALIBRATION

These are suggested concentrations for manual calibration standards.

| Concentration, mg $SO_4$/L | 0 | 5.0 | 10 | 15 | 20 | 30 | 40 |
|---|---|---|---|---|---|---|---|
| Volume (mL) Auto Calibration Standard Solution diluted as indicated with DI Water | -- | 12.5 | 25 | 37.5 | 50 | 75 | 100 |
| Final Volume, mL | 100 | 100 | 100 | 100 | 100 | 100 | 100 |

Prepare weekly.

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## 10 SAMPLE PRESERVATION AND STORAGE

Store samples at 4ºC with a maximum holding time of 28 days.

The laboratory should consult the EPA for the most current acceptable sample storage and preservation procedures.

## 11 PROCEDURE

Prepare standards and reagents as described in Sections 8 and 9. The laboratory must determine standard concentrations that are most suitable to their needs for calibration within the range of application for this method. The concentrations listed within this method are only suggestions.

Pour calibration standards, quality control solutions and samples into sample cups and reagent wedges, according to the user-defined work list.

Before running the analysis batch, check the analyzer for sufficient reagent water, fresh reaction segments, correct location of reagent wedges and a waste collection bottle in place.

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# 12 TEST PARAMETERS

| PARAMETER | AQ SETTING |
|---|---|
| Test name | SO4_40 |
| Units | mg/L |
| Decimals | 2 |
| Test type | End Point |
| Sample volume (μL) | 400 |
| Diluent location | *Lab specified* |
| Water volume (μL) | 0 |
| Number of mixes | 2 |
| Cuvette primes | 2 |
| Cuvette washes | 2 |
| Baseline on Wash | Ticked |
| Reaction time (seconds) | 270 |
| Wavelength (nm) | 405 |
| Polynomial order | 2 (or 3) |
| Number of reagents | 1 |
| 1. Turbidimetric Reagent (μL) | 200 |
| Advanced Parameters | ☒ Extra Debubbling Action |

**DEFINITIONS:**

| | |
|---|---|
| Test name: | Name which appears on the final report |
| Units: | Concentration units appearing against the reported result |
| Decimals: | Number of decimal places in the reported result |
| Test type: | The type of test the software will perform |
| Sample volume: | Volume (in microliters) of the sample used in the test |
| Water volume: | Volume (in microliters) of water from the water reservoir used in the test |
| Number of mixes: | Number of mixes of the sample + reagent in the reaction well |
| Cuvette primes: | Number of times the cuvette is flushed with reagent (no sample) prior to sampling |
| Cuvette washes: | Number of washes of the cuvette between aspirations of finished reaction liquid |
| Baseline on Wash: | Water baseline measured between aspirations to provide drift correction |
| Reaction time: | Time (in seconds) between addition of sample + reagent and measurement of the reaction mixture |
| Wavelength: | Interference filter wavelength to be used in the test |
| Polynomial order: | Polynomial order to be used when fitting the standard points to the standard graph (1 = linear) |
| Number of reagents: | Number of reagents defined for the test |
| Reagent volume: | Volume (in microliters) of the reagent used in the test |

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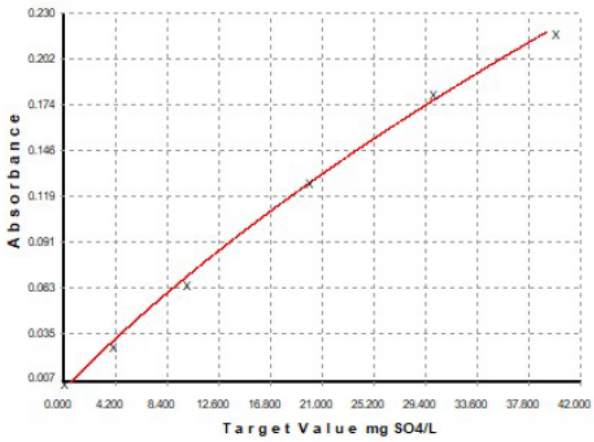
# 13 STANDARDIZATION

**Calibration Chart**

| Type | Absorbance | Calc mg SO4/L | Target mg SO4/L | % Error | | |
|---|---|---|---|---|---|---|
| S1 | 0.0074 | 0.543 | 0.000 | | Polynomial order : | 2 |
| S90 | 0.0302 | 3.712 | 4.000 | -7.19 | Correlation Coefficient : | 0.9990 |
| S91 | 0.0668 | 9.295 | 10.000 | -7.05 | RSE (%) : | 7.59 |
| S92 | 0.1283 | 20.106 | 20.000 | 0.53 | Carryover(%) : | -0.0 |
| S93 | 0.1819 | 30.997 | 30.000 | 3.32 | Calibration equation: | y = cx^2 + bx + a |
| S94 | 0.2190 | 39.347 | 40.000 | -1.63 | y =: | Concentration mg SO4/L |
| S0 | 0.0073 | 0.533 | 0.000 | | x =: | Measured absorbance |
| | | | | | a =: | 1.296069E+02 |
| | | | | | b =: | 2.373872E+02 |
| | | | | | c =: | 0.000000E+00 |
| | | | | | Date & Time : | 2023-01-18 12:03:57 |

**Calibration Graph**




Max. Blank Absorbance (S1): 0.02 AU

Sensitivity Range (40 mg $SO_4$/L): 0.18 – 0.26 AU

**DEFINITIONS:**

Abs. Range: Difference between the absorbance of the highest standard and the absorbance of the blank

Variance: Calculated as follows:

$$\text{Variance} = \frac{\sum(\text{Deviations, i.e., errors})^2}{\sum(\text{Observed values})^2} \times 100$$

Carryover (%): Calculated as follows:

$$\text{Carryover (\%)} = \frac{(A - B)}{C} \times 100\%$$

Where, A = absorbance of carryover blank (S0)

B = absorbance of blank (S1)

C = calibration absorbance range

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# 14 PERFORMANCE VALIDATION DATA

## 14.1 DETECTION LIMIT STUDY

Replicates of blanks of reagent water***

| | Concentration (mg $SO_4$/L) | | |
|---|---|---|---|
| | 0.32 | -0.42 | 0.21 |
| | 0.31 | -0.44 | 0.12 |
| | 0.46 | -0.50 | 0.13 |
| | 0.39 | -0.44 | 0.15 |
| | 0.35 | -0.44 | 0.13 |
| | 0.46 | -0.47 | 0.15 |
| | 0.37 | -0.44 | 0.21 |
| | 0.44 | -0.41 | 0.21 |
| | 0.37 | -0.43 | 0.19 |
| | 0.41 | -0.46 | 0.11 |
| Statistics | | | |
| Mean | 0.41 | -0.45 | 0.16 |
| Std. Deviation | 0.04 | 0.03 | 0.04 |
| MDL* | 0.12 | 0.07 | 0.11 |
| RL/LOQ*+ | 0.44 | 0.26 | 0.38 |

## 14.2 PERFORMANCE STUDY

Replicates of 4.0 mg $SO_4$/L standard in reagent water***

| | Concentration (mg $SO_4$/L) | | |
|---|---|---|---|
| | 3.92 | 4.04 | 4.38 |
| | 4.04 | 3.89 | 4.27 |
| | 3.99 | 3.99 | 4.38 |
| | 3.93 | 3.92 | 4.24 |
| | 3.94 | 3.90 | 4.21 |
| | 3.93 | 3.90 | 4.30 |
| | 3.98 | 3.97 | 4.11 |
| | 3.90 | 3.97 | 4.15 |
| | 3.87 | 3.98 | 4.23 |
| | 3.90 | 3.97 | 4.26 |
| Statistics | | | |
| Mean | 3.94 | 3.95 | 4.25 |
| Std. Deviation | 0.05 | 0.05 | 0.09 |
| % RSD** | 1.3% | 1.3% | 2.1% |
| % Recovery | 98.5% | 98.8% | 106.3% |

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## 14.3 PRECISION STUDY

Replicates of 20 mg $SO_4$/L standard in water***

Concentration (mg $SO_4$/L)

| Statistics | | | |
|---|---|---|---|
| Mean | 19.77 | 19.72 | 20.73 |
| % Recovery | 98.9 | 98.6 | 103.6 |
| Std. Deviation | 0.32 | 0.24 | 0.49 |
| % RSD** | 1.6 | 1.2 | 2.3 |
| Maximum | 20.38 | 20.19 | 21.75 |
| Minimum | 19.40 | 19.42 | 20.10 |

Replicates of 40 mg $SO_4$/L standard in water***

Concentration (mg $SO_4$/L)

| Statistics | | | |
|---|---|---|---|
| Mean | 38.30 | 38.71 | 39.56 |
| % Recovery | 95.7 | 96.8 | 98.9 |
| Std. Deviation | 0.54 | 0.31 | 0.60 |
| % RSD** | 1.4 | 0.8 | 1.5 |
| Maximum | 39.21 | 39.21 | 40.88 |
| Minimum | 37.36 | 38.36 | 38.74 |

Data collected on AQ400.

* MDL is calculated by multiplying the Standard Deviation by the Student-t value for the number of replicates run.

*+LOQ is calculated as 10x the Standard Deviation

** %RSD is calculated by dividing the Standard Deviation by the mean, then multiplying by 100.

*** To prepare these standards, refer to Section 9.

# 15 REVISIONS

| | | |
|---|---|---|
| March 2024 | Rev A | Method created. |

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